In [10]:
```
# Import required libraries
import numpy as np
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Breast Cancer dataset
data = load_breast_cancer()

# Create DataFrame
df = pd.DataFrame(data.data, columns=data.feature_names)

# Add target column
df['target'] = data.target

print("First 5 Rows of Dataset:")
print(df.head())

# Features and target
X = data.data
y = data.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create SVM model with Sigmoid (Logistic) Kernel
svm_model = SVC(kernel='sigmoid')

# Train the model
svm_model.fit(X_train, y_train)

# Predict on test data
y_pred = svm_model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
First 5 Rows of Dataset:
   mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0        17.99         10.38          122.80     1001.0          0.11840
1        20.57         17.77          132.90     1326.0          0.08474
2        19.69         21.25          130.00     1203.0          0.10960
3        11.42         20.38           77.58      386.1          0.14250
4        20.29         14.34          135.10     1297.0          0.10030

   mean compactness  mean concavity  mean concave points  mean symmetry  \
0           0.27760          0.3001              0.14710         0.2419
1           0.07864          0.0869              0.07017         0.1812
2           0.15990          0.1974              0.12790         0.2069
3           0.28390          0.2414              0.10520         0.2597
4           0.13280          0.1980              0.10430         0.1809

   mean fractal dimension  ...  worst texture  worst perimeter  worst area  \
0                 0.07871  ...          17.33           184.60      2019.0
1                 0.05667  ...          23.41           158.80      1956.0
2                 0.05999  ...          25.53           152.50      1709.0
3                 0.09744  ...          26.50            98.87       567.7
4                 0.05883  ...          16.67           152.20      1575.0

   worst smoothness  worst compactness  worst concavity  worst concave points  \
0            0.1622             0.6656           0.7119                0.2654
1            0.1238             0.1866           0.2416                0.1860
2            0.1444             0.4245           0.4504                0.2430
3            0.2098             0.8663           0.6869                0.2575
4            0.1374             0.2050           0.4000                0.1625

   worst symmetry  worst fractal dimension  target
0          0.4601                  0.11890       0
1          0.2750                  0.08902       0
2          0.3613                  0.08758       0
3          0.6638                  0.17300       0
4          0.2364                  0.07678       0

[5 rows x 31 columns]

Accuracy: 0.4649122807017544

Confusion Matrix:
[[ 6 37]
 [24 47]]

Classification Report:
              precision    recall  f1-score   support

           0       0.20      0.14      0.16        43
           1       0.56      0.66      0.61        71

    accuracy                           0.46       114
   macro avg       0.38      0.40      0.39       114
weighted avg       0.42      0.46      0.44       114
```

In [ ]: